

MOORED BUOY 22001, Japan

WMO: 992300

Lat: 28.300N

Lon: 126.000E

Elev: 0

StdP: 14.70

Time Zone: 8.00 (E08)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	49.0	50.5	35.1	30.0	51.2	36.5	31.7	52.2	38.7	56.0	36.3	55.4	25.0	0	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.5	86.3	80.7	85.9	80.7	85.4	80.4	82.6	85.1	81.9	84.6	81.4	84.3	13.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.8	165.5	84.6	81.0	161.2	84.1	80.6	158.6	83.9	46.5	85.2	45.8	84.9	45.2	84.6	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 45.3	35.3	30.8	46.4	87.2	2.5	0.9	44.6	87.9	43.1	88.4	41.7	88.9	39.9	89.5		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	71.2	59.3	58.2	62.0	66.8	72.4	78.1	83.1	83.3	80.5	75.4	70.3	63.7
	DBStd	9.58	4.96	5.52	5.19	5.07	3.75	3.25	1.81	1.58	2.56	3.25	4.13	4.86
	HDD50	3	1	2	0	0	0	0	0	0	0	0	0	0
	HDD65	641	185	199	122	40	3	0	0	0	0	0	7	85
	CDD50	7730	290	232	372	505	695	844	1026	1031	915	789	608	423
	CDD65	2894	10	10	29	94	233	394	561	566	465	324	165	43
	CDH74	24589	0	0	5	120	682	3154	6806	6989	4771	1714	338	10
	CDH80	6549	0	0	0	0	10	453	2407	2551	1060	67	1	0
Wind	WSAvg	16.5	18.9	18.4	17.1	14.9	13.5	14.4	15.5	16.6	17.2	17.7	16.9	17.2
Precipitation	PrecAvg	76.2	2.9	3.4	5.4	7.4	8.9	13.5	5.2	7.8	9.6	5.1	4.2	3.0
	PrecMax	101.5	6.1	9.5	14.5	16.2	15.4	20.6	17.3	18.2	19.0	11.3	10.3	6.8
	PrecMin	54.2	2.1	1.9	1.4	3.6	2.8	3.4	0.5	0.6	1.1	1.5	2.0	1.8
	PrecStd	12.2	1.1	2.1	2.6	2.5	3.3	4.3	4.2	4.6	4.4	2.6	1.9	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.5	72.6	74.8	77.9	81.3	84.5	87.1	87.2	86.1	82.8	80.0	75.3
		MCWB	68.7	69.5	72.2	75.1	78.3	80.5	81.0	80.5	80.6	78.1	72.3	70.7
	2%	DB	70.7	71.0	73.0	76.5	79.4	83.6	86.3	86.4	85.2	81.6	78.7	73.8
		MCWB	65.8	67.4	70.9	73.7	76.6	80.5	81.0	80.5	80.0	77.2	73.6	68.9
	5%	DB	68.9	69.2	71.8	75.6	78.2	83.0	85.9	86.1	84.5	80.7	77.2	72.4
		MCWB	63.5	65.3	69.8	73.2	75.0	80.3	80.9	80.6	79.4	76.2	72.5	67.1
	10%	DB	67.0	67.0	70.2	74.2	77.2	82.4	85.4	85.5	83.8	79.9	76.0	70.9
		MCWB	62.0	63.0	67.5	71.6	73.7	80.1	80.6	80.2	78.8	74.9	71.0	65.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.2	69.9	72.7	76.2	78.8	82.5	83.7	82.7	81.9	78.4	75.0	73.1
		MCDB	72.1	71.0	73.9	77.6	79.8	83.6	85.6	85.3	84.7	81.3	77.2	74.5
	2%	WB	67.8	66.9	71.3	75.1	77.4	81.6	82.9	81.9	81.4	77.6	73.8	70.1
		MCDB	70.2	68.9	72.6	76.2	78.7	83.0	85.3	85.1	84.4	80.7	76.4	73.4
	5%	WB	65.3	64.7	69.5	74.2	76.3	80.9	82.4	81.6	80.6	76.1	72.7	67.6
		MCDB	68.2	67.6	71.1	75.5	77.7	82.6	85.1	84.8	84.0	79.2	75.6	71.2
	10%	WB	63.1	62.4	66.7	73.0	75.2	80.5	81.8	81.0	79.8	74.2	71.4	65.4
		MCDB	66.8	65.9	69.0	74.6	76.8	82.3	84.7	84.4	83.3	77.7	74.7	69.6
Mean Daily Temperature Range	5% DB	MDBR	5.3	5.4	6.2	5.9	4.6	3.8	2.6	2.5	2.8	3.1	4.1	4.4
		MCDBR	8.1	7.3	7.3	6.2	5.2	3.3	2.3	2.5	2.9	3.1	4.6	5.8
		MCWBR	8.6	8.5	9.0	7.0	4.8	1.9	2.1	2.2	2.9	3.4	4.7	6.5
	5% WB	MCDBR	8.2	7.7	8.9	6.4	5.2	2.8	2.5	2.5	2.4	3.2	5.7	5.7
		MCWBR	9.7	9.4	9.8	7.5	5.5	2.3	2.5	2.7	2.7	3.7	6.0	7.1
Clear-Sky Solar Irradiance	taub	0.490	0.532	0.591	0.599	0.551	0.531	0.502	0.508	0.504	0.494	0.485	0.481	
	taud	1.982	1.905	1.793	1.834	2.018	2.138	2.238	2.217	2.186	2.128	2.085	2.012	
	Ebn at Noon	230	233	227	232	243	246	253	251	248	241	231	226	
	Edh at Noon	49	57	68	67	56	49	45	45	45	45	44	46	
All-Sky Solar Radiation	RadAvg	627	788	1086	1313	1494	1486	2066	1887	1570	1237	838	650	
	RadStd	114	159	118	137	197	155	199	124	140	102	83	78	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page